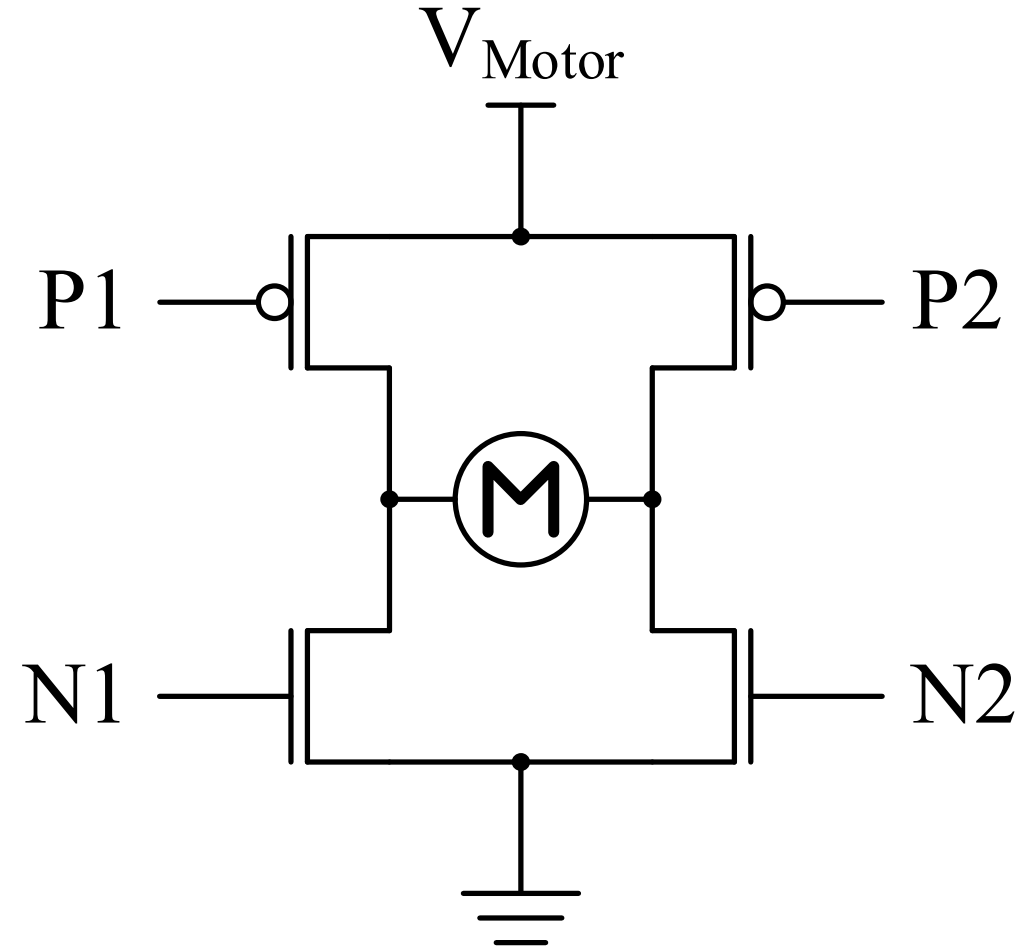


2.4.3 – Transistor Circuits

Digital: H-Bridge Circuits

- Steers current in either direction
- Complete with '1' = V_{Motor} , '0' = GND

Mode	P1	P2	N1	N2
Forward				
Reverse				
Coast				
Brake				
Short Circuit				
Short Circuit				

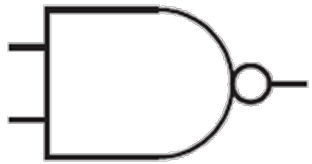


- Can be used whenever bidirectional current is required

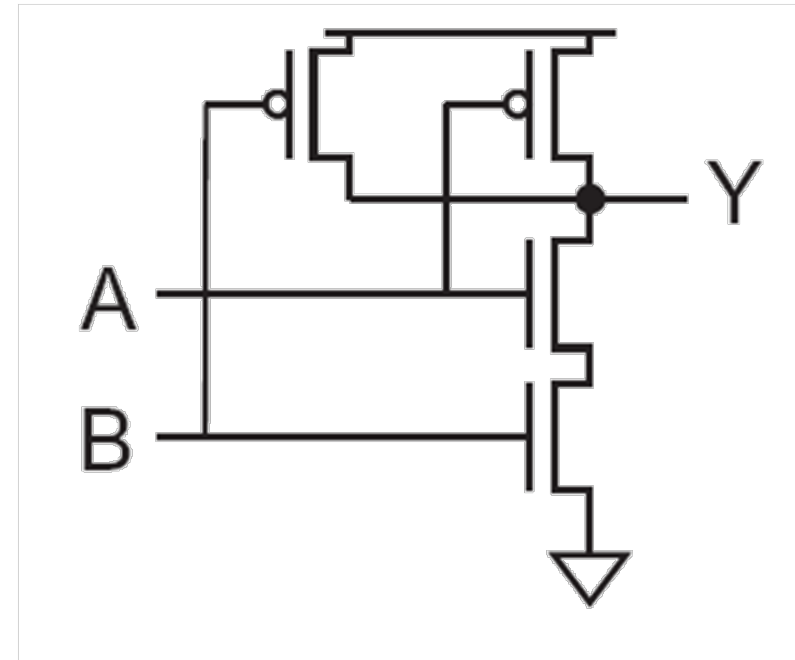


Digital CMOS

- PMOS pulls-up to output '1'
- NMOS pulls down to output '0'
- E.g., 2-input NAND gate

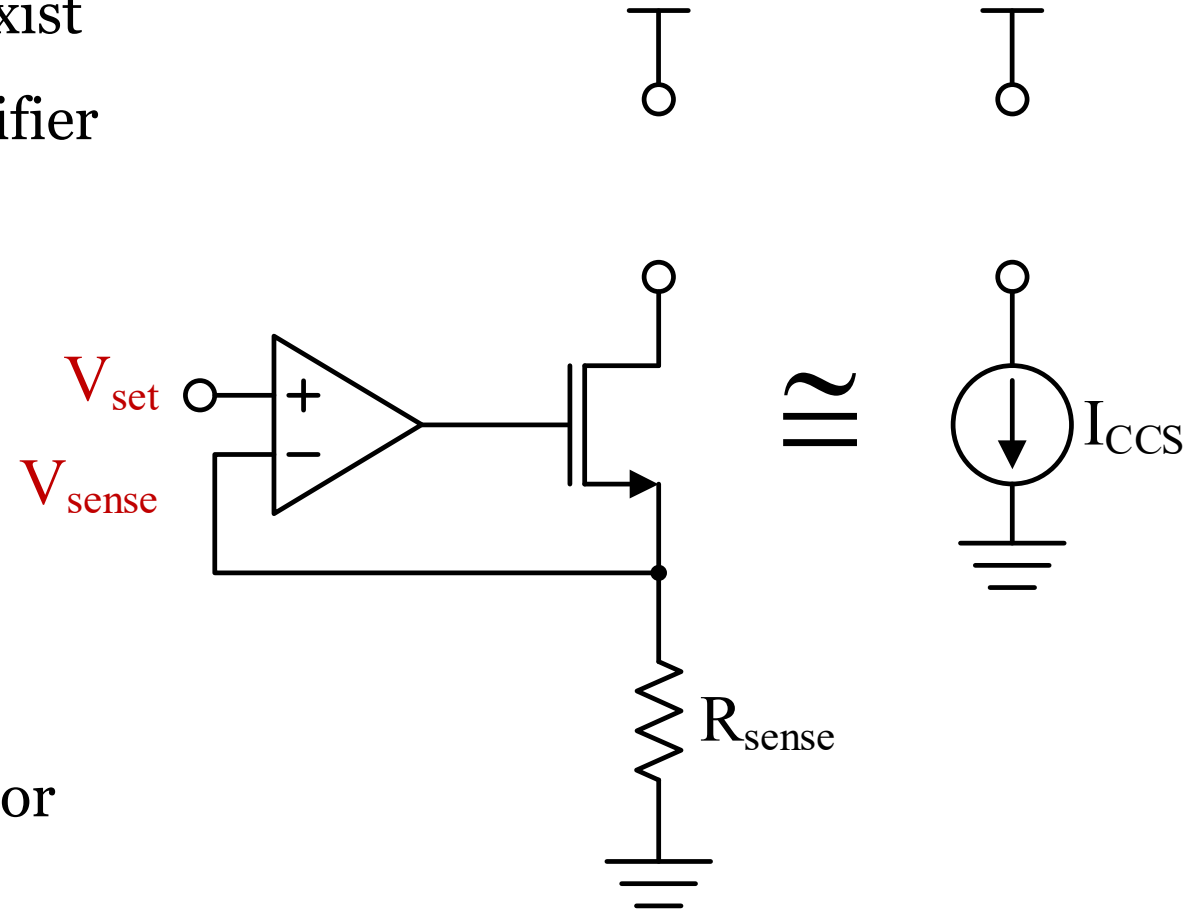


A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0



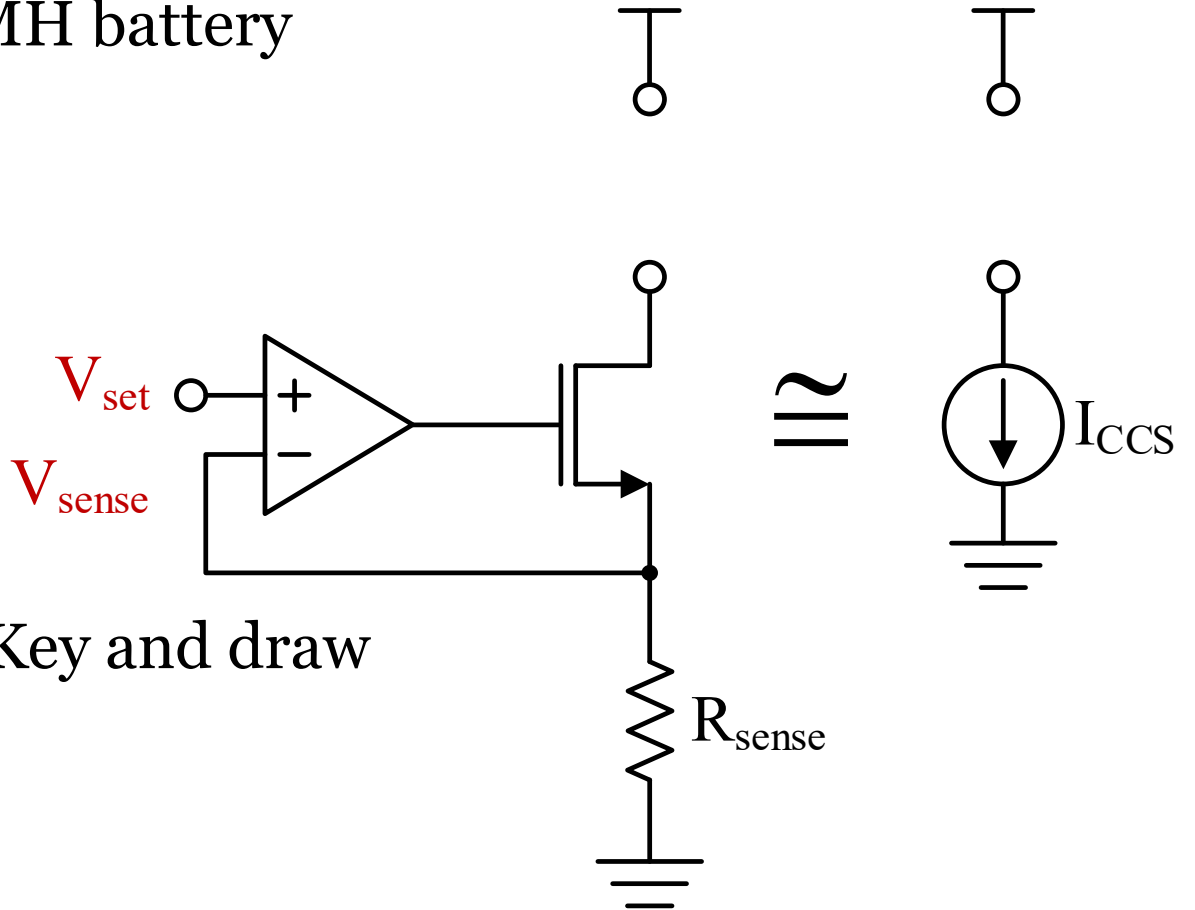
MOSFETs as Variable Resistors: Controlled Current Source

- Constant Current source does not actually exist
- Make with a current-sensing feedback amplifier
 - Sense current using voltage across a low-value resistor (to save on I^2R losses)
 - Control MOSFET gate voltage to act as variable resistor
 - Negative feedback to maintain desired current
- Called OTA: Operational Transconductance Amplifier [A/V]
- Used in: Battery charging, LED drivers, motor torque control, electrochemical reactions



Controlled Current Source Example

- Design this circuit to charge a 14.4 V NiMH battery
- Given:
 - I_{CCS} from 0 to 2 A
 - An MCU DAC can generate V_{set} from 0 to 3.3 V
 - $V_{DD} = 18\text{ V}$
- Choose R_{sense} and a MOSFET from Digi-Key and draw the completed circuit.

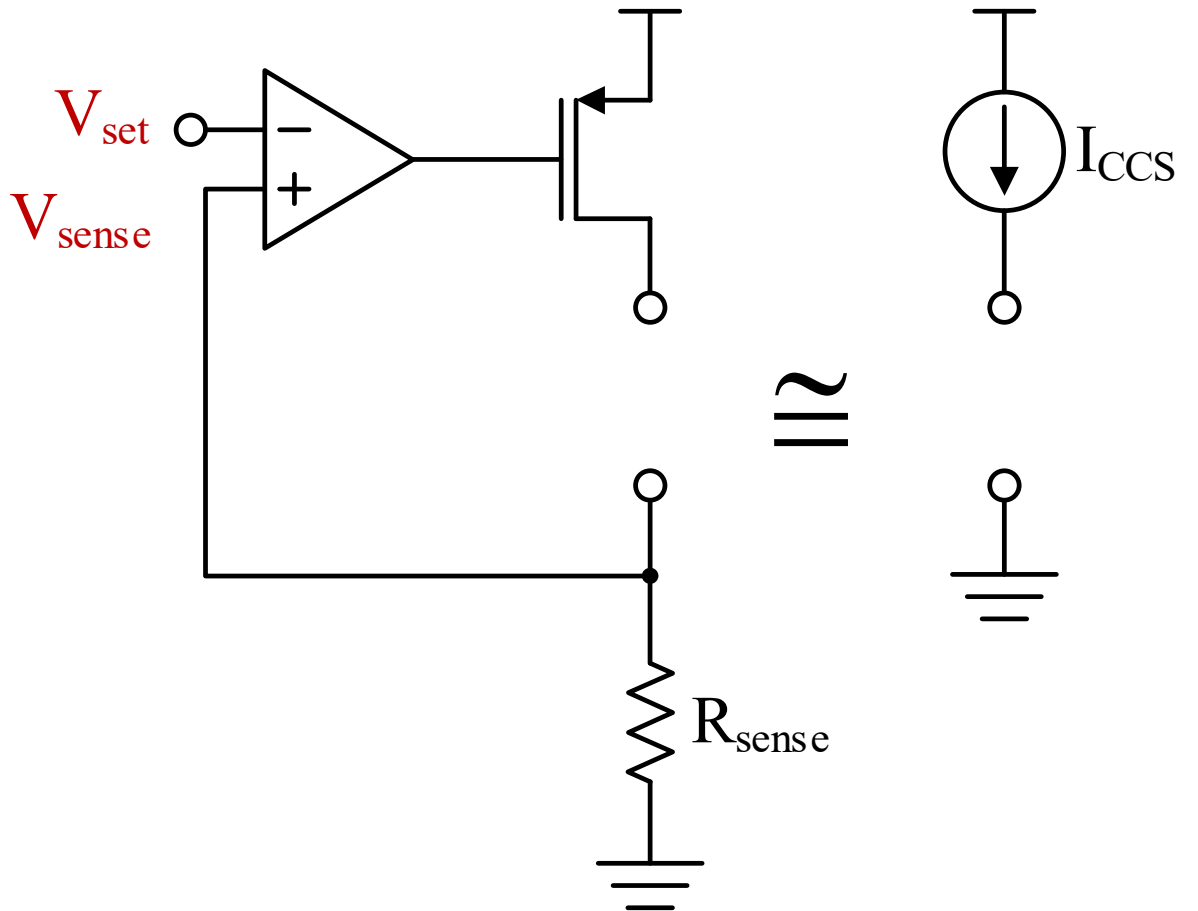


Controlled Current Source Example

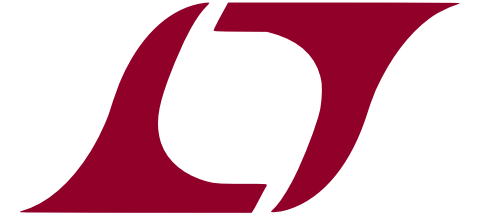


PMOS Controlled Current Source

- Previous NMOS circuit sinks current to GND (Current Sink)
 - Recall NMOS good at “pulling down” to GND but not “pulling up” to VDD
 - Also note exposed terminal connected to VDD (possible safety issue)
- Current Source requires PMOS
 - PMOS inverted action flips feedback path (still negative feedback)
 - Exposed terminal is ~GND
 - PMOS less efficient than NMOS, but still usually okay



LTSpice Modeling Challenge



- Implement a battery charging circuit
- Start with the NMOS Controlled Current Source example
 - Modify it for a battery of your choosing (find the charging requirements!)
- Model the battery as a large capacitor (voltage will drop with discharge):
 - $C_{eq} = 3600B / (V_{max} - V_{min})$
 - C_{eq} is the equivalent capacitance in F
 B is the battery capacity in A·h
 $V_{max} - V_{min}$ is the difference between charged and discharged battery voltages in V
 - E.g., a 14.4 V NiMH battery might have $B = 3$ Ah, $V_{max} - V_{min} = 17.4 - 12.0 = 5.4$ V, giving $C_{eq} = 2000$ F
 - You will need to set the capacitor's initial voltage for the simulation
- Try improving it: Can you get the battery negative terminal grounded? Can you detect and protect against a short circuit? Can you implement a realistic charging cycle recommended by the battery manufacturer? What other improvements can you make?

